

Lab 2.1: Implementation of DFA to identify strings having a's and b's in multiples of 3 in sequence

Background and Requirements

In **Formal Language Theory** and **Automata Theory**, a Deterministic Finite Automaton (DFA) is a mathematical model used to recognize whether a string belongs to a particular language. A DFA processes an input string one character at a time and moves through a finite number of states according to predefined transition rules. After reading the entire input, the string is accepted if the automaton ends in an accepting state; otherwise, it is rejected.

DFAs are widely used in compiler design, lexical analysis, pattern recognition, text processing, and digital system design because they provide an efficient method for validating strings against specific rules.

The task is to accept a string consisting of the symbols 'a' and 'b' and determine whether the string has a's and b's in multiples of 3 in sequence.

Source Code

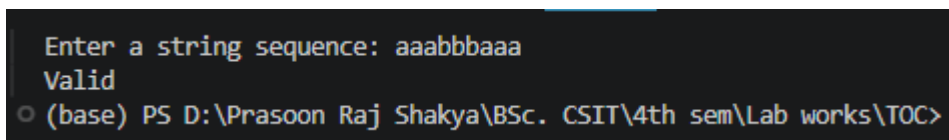
```
alphabet = {"a", "b"}
s = input("Enter a string sequence: ")
state = "A"

for ch in s:
    if ch not in alphabet:
        state = "dead"
        break
    match state:
        case "A":
            state = "B" if ch == "a" else "E"
        case "B":
            if ch == "a":
                state = "C"
            else:
                state = "dead"
                break
        case "C":
            if ch == "a":
                state = "D"
            else:
                state = "dead"
                break
        case "D":
            state = "B" if ch == "a" else "E"
        case "E":
            if ch == "b":
                state = "F"
            else:
                state = "dead"
                break
        case "F":
```

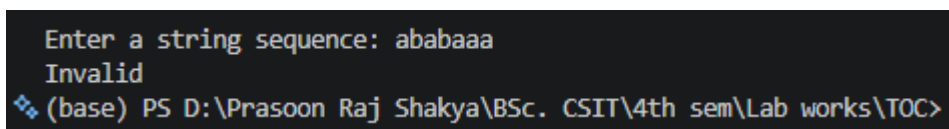
```
    if ch == "b":
        state = "D"
    else:
        state = "dead"
        break

if state == "D":
    print("Valid")
else:
    print("Invalid")
```

Output/Observations

A terminal window with a dark background. The prompt is a blue circle followed by the text "(base) PS D:\Prasoon Raj Shakya\BSc. CSIT\4th sem\Lab works\TOC>". The user has entered "aaabbbaaa" and the program has responded with "Valid".

```
Enter a string sequence: aaabbbaaa
Valid
(base) PS D:\Prasoon Raj Shakya\BSc. CSIT\4th sem\Lab works\TOC>
```

A terminal window with a dark background. The prompt is a blue diamond followed by the text "(base) PS D:\Prasoon Raj Shakya\BSc. CSIT\4th sem\Lab works\TOC>". The user has entered "ababaaa" and the program has responded with "Invalid".

```
Enter a string sequence: ababaaa
Invalid
(base) PS D:\Prasoon Raj Shakya\BSc. CSIT\4th sem\Lab works\TOC>
```

Conclusion

Through this lab, an understanding of Deterministic Finite Automata (DFA) and their role in string validation was developed. The concepts of states, transitions, input alphabets, and accepting states were explored through practical implementation in Python. The lab demonstrated how strings can be validated against predefined rules using state-based processing, thereby reinforcing the fundamental principles of automata theory and pattern recognition.